

A Comparison of Elite and Sub-elite Match-Play Running Performance of Gaelic Football Players

Sam Donnelly,¹ Kieran Collins,¹ Con Burns,² Cian O'Neill,² and Shane Mangan^{1,2}

¹Gaelic Sports Research Centre, Technological University Dublin, Tallaght, Dublin, Ireland; and ²Department of Sport, Leisure & Childhood Studies, Munster Technological University, Munster, Ireland

Abstract

Donnelly, S, Collins, K, Burns, C, O'Neill, C, and Mangan, S. A comparison of elite and sub-elite match-play running performance of Gaelic football players. *J Strength Cond Res* 38(5): 912–923, 2024—This research aimed to examine differences in match-play running performance between elite and sub-elite Gaelic football players. Global Positioning System devices were used to record the match-play running performance of sub-elite and elite players during the 2018 and 2019 seasons, respectively. In total, 783 samples were collected from 31 sub-elite and 30 elite players, from 1 elite and 1 sub-elite team. Comparisons were made on a team and positional level. Statistical significance was accepted at $p \leq 0.05$. Significant positional differences were present between all positions with varying effect sizes (ESs) and p -values. Effect size was used to determine the magnitude of statistical difference. On a team level, the elite covered significantly more relative distance ($p = 0.021$, $ES = 0.007$) and elicited significantly higher average speed ($p = 0.023$, $ES = 0.007$) in quarters 2, 3, and 4 than the sub-elite. The elite covered significantly more relative distance high-intensity running ($17\text{--}21.9 \text{ km}\cdot\text{h}^{-1}$; $p = 0.002$, $ES = 0.012$) in quarter 3 and performed significantly more power events per minute ($p = 0.006$, $ES = 0.009$) in quarter 4. In quarter 1, the sub-elite performed significantly more power events per minute ($p = 0.006$, $ES = 0.009$), relative high-speed running ($\geq 17 \text{ km}\cdot\text{h}^{-1}$; $p = 0.011$, $ES = 0.009$), and relative distance high-intensity running ($17\text{--}21.9 \text{ km}\cdot\text{h}^{-1}$; $p = 0.002$, $ES = 0.012$). Power events were defined as the estimation of the number of events that depend on anaerobic processes. The present study indicates that elite players possess superior conditioning, ability to read match-play, pacing, and anaerobic capacity. Sub-elite teams may use the present study to inform their training to potentially improve conditioning, pacing, and anaerobic capacity.

Key Words: team sport, GPS, match analysis

Introduction

Gaelic football is an intermittent running-based sport indigenous to Ireland and is governed by the Gaelic Athletic Association (GAA) (17,21,34). It includes skills and actions such as kicking, blocking, passing, tackling, marking, catching, handling, long-distance kicks, soloing (kicking the ball to oneself and collecting it before bouncing), and intercepting passes (27). In elite and sub-elite Gaelic football, each game consists of 2 opposing teams containing 14 outfield players, 3 full backs, 3 half backs, 2 midfielders, 3 half forwards, 3 full forwards, and 1 goalkeeper. Club-level competition is considered to be sub-elite, and the best of these sub-elite players are selected to compete at the intercounty level, which are considered to be elite (7,13,21,26). Thus, all players compete at the sub-elite level; however, only the best sub-elite players compete at the elite level. Elite matches consist of 35-minute halves separated by a 15-minute half-time interval. Sub-elite matches consist of two 30-minute halves separated by a 10-minute interval. Matches may also have additional time to account for stoppages (19).

Matches are played on a natural or synthetic field, 130–145 m long and 80–90 m wide, with H-shaped goalposts with a crossbar 2.5 m from the ground at each end of the field. A team is awarded 1 point for playing the ball over the crossbar and 3 points for playing the ball under the crossbar (30). The objective of the sport is to score more points than the opposing team (30).

Global Positioning System (GPS) units are used in practice and within literature in Gaelic football to monitor athletic and running performance (19,20,22). Running performance has been used as a key performance indicator and is influenced by factors including position, time of season, match-quarter, and opposition quality (6,14,16,21). Within team sports, running speed is regularly categorized in speed zones (8). The most common speed zones reported throughout Gaelic football literature regarding running performance are categorized as follows: standing and walking ($0\text{--}6.9 \text{ km}\cdot\text{h}^{-1}$), jogging ($7\text{--}11.9 \text{ km}\cdot\text{h}^{-1}$), cruising or striding ($12\text{--}16.9 \text{ km}\cdot\text{h}^{-1}$), high-intensity running ($17\text{--}21.9 \text{ km}\cdot\text{h}^{-1}$), and sprinting ($>22 \text{ km}\cdot\text{h}^{-1}$) (14,19,20,22,28). It has been reported that elite Gaelic footballers cover a relative total distance of $116 \pm 21 \text{ m}\cdot\text{min}^{-1}$ during match-play (16). In elite-level Gaelic football, 16% of match-play running is classified as high-intensity ($>17 \text{ km}\cdot\text{h}^{-1}$), covering $25 \pm 9 \text{ m}\cdot\text{min}^{-1}$ of high-speed running, with midfielders covering the greatest total and high-speed distance (16,23). It has also been reported that elite players perform $524 \pm 190 \text{ m}$ of very-high-speed running ($\geq 22 \text{ km}\cdot\text{h}^{-1}$) throughout the duration of a match (32). To the author's knowledge, very-high-speed running has not been reported in relative terms ($\text{m}\cdot\text{min}^{-1}$) within an elite Gaelic football cohort; however, the present study will provide data regarding this metric.

In sub-elite Gaelic football, players will typically cover a relative total distance of $112 \pm 18 \text{ m}\cdot\text{min}^{-1}$, with half backs, half forwards, and midfielders running covering the most distance (19). Sub-elite players also cover $42.3 \pm 3.5 \text{ m}\cdot\text{min}^{-1}$ walking ($0\text{--}6.9 \text{ km}\cdot\text{h}^{-1}$), $23.1 \pm 5.0 \text{ m}\cdot\text{min}^{-1}$ jogging ($7\text{--}11.9 \text{ km}\cdot\text{h}^{-1}$),

Address correspondence to Sam Donnelly, donnellysam72@gmail.com.

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$24.6 \pm 8.8 \text{ m} \cdot \text{min}^{-1}$ cruising or striding ($12\text{--}16.9 \text{ km} \cdot \text{h}^{-1}$), $14.9 \pm 5.5 \text{ m} \cdot \text{min}^{-1}$ while high-intensity running ($17\text{--}21.9 \text{ km} \cdot \text{h}^{-1}$), and $6.7 \pm 2.7 \text{ m} \cdot \text{min}^{-1}$ sprinting ($>22 \text{ km} \cdot \text{h}^{-1}$) (19).

Previous research has examined performance indicators and running performance in elite Gaelic footballers, but few have examined these variables in sub-elite cohorts (19). To date, only one study has examined sub-elite running performance in Gaelic football (19). The study by Mangan et al. (19) examined division 1 and division 2 sub-elite teams and found minimal differences in technical or running performance between divisions. Because of discrepancies in methodological approaches and devices used in the domain, it has not been possible to compare data sets from different, yet related, studies because this would likely lead to error and unreliable results. As such, the present investigation comparing running performance between elite and sub-elite levels within Gaelic football is warranted to draw reliable comparisons to understand in which aspects of running performance elite-level teams differ in comparison with sub-elite teams. Additionally, previous studies comparing elite and sub-elite Gaelic players have not examined running performance and may be outdated because of progressions within the sport in training practices and tactical play, which further highlights the need for the present study (13,26). One recent study compared elite and sub-elite match-play performance in hurling (38). Hurling is also a national game in Ireland, has the same pitch size, goal size, and number of players as Gaelic football but is played with a stick (i.e., a hurl) and a smaller ball (i.e., a sliotar) (19). Young et al. (38) found that elite hurlers covered a greater total distance ($118 \pm 9 \text{ m} \cdot \text{min}^{-1}$) than the sub-elite hurlers ($93 \pm 16 \text{ m} \cdot \text{min}^{-1}$), but the sub-elite hurlers performed more relative high-speed running ($3.3 \pm 0.4 \text{ m} \cdot \text{min}^{-1}$ vs. $2.9 \pm 1.1 \text{ m} \cdot \text{min}^{-1}$). The differences in running performance variables between competition levels in hurling, presented by Young et al. (38), further strengthens the rationale for the examination of running performance variables in Gaelic football because of the similarities between the games (i.e., pitch-size, goal-size, and number of players). Additionally, of over 370,000 players registered with the GAA, only 0.3% of individuals who participate in Gaelic football play at the elite level (18,19). This disproportionately large participation in sub-elite compared with elite Gaelic football presents a need for more research comparing elite and sub-elite Gaelic footballers, to allow for sub-elite players to adjust training and tactics within match-play and to increase their standard of play and performance.

The aim of this study was to identify the differences in running performance between elite and sub-elite Gaelic football players and potentially set forth reasoning for these differences. Players are scouted for elite-level teams during sub-elite competition. As such, understanding elite-level match-play running demands and attempting to meet these demands may lead to an increased standard of play at the sub-elite level and may increase the likelihood of many sub-elite players progressing to the elite level (38). It is hypothesized that given the differences presented within previous research in hurling, that differences in running performance will be present in the current investigation and because of the increased level of preparation that occurs among elite-level players, they will possess superior conditioning (37,38).

Methods

Experimental Approach to the Problem

Full or competitive match-play running performance metrics totals and metrics within each respective quarter were compared

between the elite and sub-elite cohort using GPS technology (GPX LT 18 Hz; Exelio, Udine, Italy). A total of 783 individual samples were collected and data were divided according to playing level (i.e., elite or sub-elite) and positional line played on the field in each individual match. All matches occurred between 14.00 and 21.00 hours with temperatures ranging between 5 and 15.5°C and were played on a natural grass field. On average, the sub-elite matches were $65:16 \pm 5:04$ minutes long and elite matches were $76:38 \pm 3:44$ minutes. It was requested that all players partaking in the study did not perform strenuous physical activity during the 24–48 hours before each match (16). Additionally, the average number of training sessions before a match was 2–3 pitch sessions and 1 gym session. Typically, the pitch sessions occurred 2 and 4 days before a game and the gym session occurred 5 days before a game.

Subjects

Subject were 61 adult (over 18 years of age) male Gaelic football players: 31 sub-elite and 30 elite Gaelic football players. A total of 27 sub-elite and 16 elite competitive matches were analyzed, during the 2018 and 2019 seasons, respectively. The volunteers for this study included 1 intercounty elite team and 1 division 1 sub-elite club team. Before signing an institutionally approved consent document to participate in the study, all participants were briefed about the purpose, benefits, risks, and methodologies of the research. Ethics and all procedures were approved by the host institute's ethics committee before the commencement of the study.

Procedures

Running performance data were collected using GPS devices. This GPS device was found to be valid and reliable for total distance covered (-1.6 ± 0.3 , bias % $\pm 90\%$ CI; 1.9 ± 0.2 , typical error of estimate between units [TEE] $\pm 90\%$ CI), and bias range from -9.2 ± 0.3 to -6.7 ± 0.6 in sprints up to 30 m (12). This GPS device is also reliable and valid for sprint variables such as theoretical max running velocity (4.5, TEE [%], 3.1, CV%) and reliable, but not valid, for theoretical max horizontal force (14.2, TEE [%], 7.5, CV%) and theoretical max horizontal power (12.5, TEE [%], 7.4, CV%) (12). Twenty-hertz GPS devices, produced by the same manufacturer as the GPS devices used in the current study, have presented a percentage bias (90% CL), ranging from -9.3 to -6.5 , -0.2 to 1.1 , and -8.1 to -5.7 for theoretical max horizontal force, theoretical max running velocity, and max horizontal power, respectively (25). Additionally, the reliability (TE [90% CL]) of the 20 Hz is moderate for theoretical maximal horizontal force (5.6 [5.0–6.4]) and good for theoretical maximal running velocity (2.5 [2.2–2.8]) and maximal horizontal power (4.5 [4.0–5.2]) (35). The biases presented for each variable are acceptable (33).

The GPS device was enclosed in a compression vest and was situated between the scapulae, in the upper thoracic-spine of each player, to limit movement of the device (12). Each GPS device was switched on and satellite lock was established an hour prior to the beginning of each match (19). Times for the start and end of each quarter and times for substitutions were noted by the primary researcher. Match-play was analyzed by quarter because of the prevalence of this methodology within Gaelic football literature (5,14,15,21,28,31). Additionally, differences in running performance variables such as high-speed running have been found between quarters within Gaelic football research, further

validating its use within the current study (5,14,32). A quarter was defined as 50% of a half of match-play, including any additional time. Average durations of quarters analyzed within the current study can be found in Table 1. Using the GPEXE bridge software, the data were uploaded to the GPEXE Web App after each match. The files were trimmed to ensure only match-play time was included. Only complete full match data sets were included in the analysis with a total of 27 sub-elite matches and 16 elite matches analyzed. If a player were substituted, they were excluded from the data before analysis.

The variables analyzed were relative total distance ($\text{m} \cdot \text{min}^{-1}$), relative high-speed running distance ($\text{m} \cdot \text{min}^{-1}$), distance covered in each speed zone per minute ($\text{m} \cdot \text{min}^{-1}$), average speed ($\text{km} \cdot \text{h}^{-1}$), maximal velocity ($\text{km} \cdot \text{h}^{-1}$), maximum acceleration ($\text{m} \cdot \text{s}^{-2}$), maximum negative acceleration ($\text{m} \cdot \text{s}^{-2}$), power events per minute (n), and power events average time (seconds) (34). Power events are defined as the estimation of the number of events in a session, which depend on anaerobic processes (34). Power events include events such as sprints, accelerations, and changes of direction. Additionally, power events' average time was assessed as the average duration of each power event. Using the excel spreadsheet, the data were converted into relative terms (per minute, $\text{m} \cdot \text{min}^{-1}$) by dividing the distance covered by the number of minutes played within each match (19,38). This was done to account for the discrepancy in match duration between elite and sub-elite Gaelic football (70 vs. 60 minutes) to allow for accurate comparisons between running performance variables between elite and sub-elite levels. Speed zones were classified as follows: standing and walking ($0\text{--}6.9 \text{ km} \cdot \text{h}^{-1}$), jogging ($7\text{--}11.9 \text{ km} \cdot \text{h}^{-1}$), cruising or striding ($12\text{--}16.9 \text{ km} \cdot \text{h}^{-1}$), high-intensity running ($17\text{--}21.9 \text{ km} \cdot \text{h}^{-1}$), and sprinting ($>22 \text{ km} \cdot \text{h}^{-1}$) (28). Previous research regarding Gaelic football have made use of these speed zones when examining running performance (19,20,22). The data were exported into a customized Microsoft Excel Spreadsheet (Microsoft Excel V2109; Microsoft Corporation, Redmond, WA). All data were deidentified before any data analysis occurred as per appropriate ethical practice and in line with the institution's guidelines. Players were categorized by playing level and position for statistical analysis. Playing positions were divided into 5 categories: full backs, half backs, midfielders, half forwards, and full forwards (21). The present study contained 61 participants, some of whom played in differing positional lines in different matches. This resulted in 60 sub-elite and 24 elite full backs; 45 sub-elite and 35 elite full forwards; 87 sub-elite and 68 elite half backs; 52 sub-elite and 41 elite half forwards; and 36 sub-elite and 25 elite midfield files being analyzed.

Statistical Analyses

Team and positional descriptive statistics are reported as mean $\pm$ SD and are supported by the coefficient of variation percentage (CV%). Statistical analysis was performed using SPSS (Statistical Package for the Social Sciences V26.0; IBM Corp., Armonk, NY). Normality of the distribution of the data was tested using the

Shapiro-Wilk's test, whereas sphericity was assessed using Mauchly's test. Statistical significance was accepted at $\alpha \leq 0.05$. A series of 1-way repeated measure ANOVAs were performed considering factor for group (2 levels: elite and sub-elite) and for time (4 levels: quarter 1–quarter 4), analyzing each variable. Bonferroni post hoc correction was used to calculate simple main effects when an interaction was present. Partial eta squared value (η_p^2) was used to measure effect size (ES) for each 1-way repeated measure ANOVAs (4). Partial eta-squared values were interpreted as null (<0.01), small (<0.06), moderate (<0.14), and large (>0.14) (29). A series of independent *t*-tests were performed to analyze positional differences between the 2 groups (elite and sub-elite) for each variable. Effect sizes for independent *t*-tests were calculated using Cohen's *d* and interpreted as follows: <0.2 = trivial, $0.2\text{--}0.6$ = small, $0.6\text{--}1.2$ = moderate, $1.2\text{--}2.0$ = large, and >2.0 = very large (11,31). These effect size and eta-squared values are being used because of the presence of the same or similar values within the past Gaelic football and hurling research (19–21,38).

Results

Positional Comparisons

Descriptive statistics for each metric within each position for each team are presented in Tables 2 and 3, as mean $\pm$ SD and CV%. Elite full backs covered more total relative distance ($105.16 \pm 9.23 \text{ m} \cdot \text{min}^{-1}$, 95% confidence interval (CI): $101.27\text{--}109.06 \text{ m} \cdot \text{min}^{-1}$ vs. $95.3 \pm 13.81 \text{ m} \cdot \text{min}^{-1}$, 95% CI: $91.73\text{--}98.87 \text{ m} \cdot \text{min}^{-1}$, $p \leq 0.001$, ES = -0.96 [moderate]), more relative distance jogging ($27.73 \pm 4.56 \text{ m} \cdot \text{min}^{-1}$, 95% CI: $25.8\text{--}29.65 \text{ m} \cdot \text{min}^{-1}$ vs. $19.92 \pm 5.42 \text{ m} \cdot \text{min}^{-1}$, 95% CI: $18.52\text{--}21.32 \text{ m} \cdot \text{min}^{-1}$, $p \leq 0.001$, ES = -1.38 [large]), and more cruising or striding ($23.3 \pm 4.5 \text{ m} \cdot \text{min}^{-1}$, 95% CI: $21.4\text{--}25.2 \text{ m} \cdot \text{min}^{-1}$ vs. $17.45 \pm 6.37 \text{ m} \cdot \text{min}^{-1}$, 95% CI: $15.8\text{--}19.09 \text{ m} \cdot \text{min}^{-1}$, $p \leq 0.001$, ES = -1.23 [large]) than the sub-elite full backs. Elite full backs also elicited a higher average speed ($6.32 \pm 0.56 \text{ km} \cdot \text{h}^{-1}$, 95% CI: $6.08\text{--}6.55 \text{ km} \cdot \text{h}^{-1}$ vs. $5.72 \pm 0.83 \text{ km} \cdot \text{h}^{-1}$, 95% CI: $5.5\text{--}5.93 \text{ km} \cdot \text{h}^{-1}$, $p \leq 0.001$, ES = -0.97 [moderate]) and performed more power events per minute (1.4 ± 0.22 power events, 95% CI: $1.3\text{--}1.49$ power events vs. 1.19 ± 0.3 power events, 95% CI: $1.12\text{--}1.27$ power events, $p = 0.001$, ES = -0.88 [moderate]) compared with the sub-elite full backs. However, the sub-elite full backs covered more relative distance walking ($42.63 \pm 2.94 \text{ m} \cdot \text{min}^{-1}$, 95% CI: $41.87\text{--}43.39 \text{ m} \cdot \text{min}^{-1}$ vs. $38.93 \pm 2.95 \text{ m} \cdot \text{min}^{-1}$, 95% CI: $37.68\text{--}40.18 \text{ m} \cdot \text{min}^{-1}$, $p \leq 0.001$, ES = 1.15 [large]) and sprinting ($5.23 \pm 2.1 \text{ m} \cdot \text{min}^{-1}$, 95% CI: $4.69\text{--}5.77 \text{ m} \cdot \text{min}^{-1}$ vs. $4.08 \pm 1.1 \text{ m} \cdot \text{min}^{-1}$, 95% CI: $3.61\text{--}5.54 \text{ m} \cdot \text{min}^{-1}$, $p = 0.002$, ES = 0.75 [moderate]) compared with their elite counterparts.

Elite half backs covered more relative distance jogging ($30.33 \pm 5 \text{ m} \cdot \text{min}^{-1}$, 95% CI: $29.18\text{--}31.54 \text{ m} \cdot \text{min}^{-1}$ vs. $26.1 \pm 4.09 \text{ m} \cdot \text{min}^{-1}$, 95% CI: $25.22\text{--}26.97 \text{ m} \cdot \text{min}^{-1}$, $p \leq 0.001$, ES = -0.94 [moderate]) than the sub-elite half backs. However, the sub-

Table 1
Average duration of quarters.*

Playing level	Quarter 1 (<i>n</i> = 43)	Quarter 2 (<i>n</i> = 43)	Quarter 3 (<i>n</i> = 43)	Quarter 4 (<i>n</i> = 43)
Elite	18:50 $\pm$ 01:39	19:00 $\pm$ 00:58	19:00 $\pm$ 02:12	18:44 $\pm$ 02:40
Sub-elite	16:09 $\pm$ 01:14	16:00 $\pm$ 00:46	16:32 $\pm$ 01:57	16:44 $\pm$ 01:41

*Data are presented as mean $\pm$ SD.

Table 2

Speed zone and relative distance metrics of Gaelic football players in regard to position.*†

Performance variable	Full back (n = 84)				Half back (n = 155)				Midfield (n = 61)				Half forward (n = 93)				Full forward (n = 80)			
	Elite	Sub-elite	p	ES	Elite	Sub-elite	p	ES	Elite	Sub-elite	p	ES	Elite	Sub-elite	p	ES	Elite	Sub-elite	p	ES
Standing and walking (0–6.9 km·h ^{−1}) (m·min ^{−1})	38.93 ± 2.95 (7.58%)	42.63 ± 2.94‡ (6.90%)	≤0.001	1.15 (large)	38.53 ± 3.08 (7.99%)	41.18 ± 3.31‡ (8.04%)	≤0.001	0.82 (moderate)	34.68 ± 3.70 (10.67%)	42.20 ± 2.63‡ (6.23%)	≤0.001	2.75 (very large)	36.66 ± 4.70 (12.82%)	40.80 ± 4.01‡ (9.83%)	≤0.001	0.95 (moderate)	37.58 ± 2.12 (5.64%)	41.44 ± 2.88‡ (6.95%)	≤0.001	1.51 (large)
Jogging (7–11.9 km·h ^{−1}) (m·min ^{−1})	27.73 ± 4.56 (16.44%)	19.92 ± 5.42‡ (27.21%)	≤0.001	−1.38 (large)	30.33 ± 5.00 (16.49%)	26.10 ± 4.09‡ (15.67%)	≤0.001	−0.94 (moderate)	37.69 ± 6.55 (17.38%)	27.37 ± 3.99‡ (14.58%)	≤0.001	−2.33 (very large)	30.91 ± 7.09 (22.94%)	26.97 ± 5.12‡ (18.98%)	0.004	−0.71 (moderate)	24.17 ± 5.02 (20.77%)	21.47 ± 4.02‡ (18.73%)	0.009	−0.6 (moderate)
Cruising/striding (12–16.9 km h ^{−1}) (m·min ^{−1})	23.30 ± 4.50 (19.31%)	17.45 ± 6.37‡ (36.5%)	≤0.001	−1.23 (large)	29.30 ± 7.08 (24.16%)	27.80 ± 7.48 (26.91%)	0.206	−0.21 (small)	40.89 ± 7.04 (17.22%)	28.65 ± 5.68‡ (19.83%)	≤0.001	−1.95 (very large)	33.65 ± 7.39 (21.96%)	32.08 ± 9.80 (30.55%)	0.395	−0.81 (moderate)	20.97 ± 5.23 (24.94%)	20.45 ± 3.94 (19.27%)	0.627	−0.12 (trivial)
High-intensity running (17–21.9 km·h ^{−1}) (m·min ^{−1})	11.11 ± 2.11 (18.99%)	10.17 ± 3.65 (35.89%)	0.146	−0.35 (small)	15.99 ± 3.80 (23.76%)	16.96 ± 5.07 (29.89%)	0.177	0.22 (small)	19.21 ± 3.08 (16.03%)	16.75 ± 4.26‡ (25.43%)	0.016	−0.64 (moderate)	17.23 ± 4.19 (24.31%)	18.74 ± 5.23 (27.91%)	0.137	0.31 (small)	11.30 ± 2.00 (17.7%)	11.68 ± 3.50 (29.97%)	0.549	0.14 (trivial)
Sprinting (>22 km·h ^{−1}) (m·min ^{−1})	4.08 ± 1.10 (26.96%)	5.23 ± 2.10‡ (40.15%)	0.002	0.75 (moderate)	6.53 ± 2.25 (34.46%)	7.35 ± 2.88 (39.18%)	0.055	0.31 (small)	4.23 ± 1.80 (42.55%)	7.54 ± 2.39‡ (31.7%)	≤0.001	1.52 (large)	6.80 ± 3.10 (45.59%)	6.26 ± 2.18 (34.82%)	0.348	−0.23 (small)	5.35 ± 1.26 (23.55%)	5.41 ± 2.25 (41.59%)	0.875	0.04 (trivial)
Relative distance (m·min ^{−1})	105.16 ± 9.23 (8.78%)	95.30 ± 13.81‡ (14.49%)	≤0.001	−0.96 (moderate)	120.69 ± 11.91 (9.87%)	119.38 ± 14.47 (11.87%)	0.548	−0.10 (trivial)	136.69 ± 10.73 (7.85%)	122.50 ± 10.76‡ (8.78%)	≤0.001	−1.32 (large)	125.25 ± 14.31 (11.43%)	124.85 ± 14.70 (11.77%)	0.894	−0.03 (trivial)	99.36 ± 11.23 (11.3%)	100.45 ± 10.44 (10.39%)	0.655	0.10 (trivial)

*CV% = coefficient of variation percentage.
†Data are presented as mean ± SD (CV%).
‡Significantly different (*p* < 0.05) from elite level.

elite half backs covered more relative distance walking (41.18 ± 3.31 m·min^{−1}, 95% CI: 40.47–41.88 m·min^{−1} vs. 38.53 ± 3.08 m·min^{−1}, 95% CI: 37.78–39.27 m·min^{−1}, *p* ≤ 0.001, ES = 0.82 [moderate]) than their elite counterparts. All other comparisons between elite and sub-elite half backs were found to be insignificant.

Elite midfield players covered significantly more relative total distance (136.69 ± 10.73 m·min^{−1}, 95% CI: 132.26–141.12 m·min^{−1} vs. 122.5 ± 10.76 m·min^{−1}, 95% CI: 118.86–126.15 m·min^{−1}, *p* ≤ 0.001, ES = −1.32 [large]), more relative distance jogging (37.69 ± 6.55 m·min^{−1}, 95% CI: 34.99–40.4 m·min^{−1} vs. 27.37 ± 3.99 m·min^{−1}, 95% CI: 26.02–28.72 m·min^{−1}, *p* ≤ 0.001, ES = −2.33 [very large]), cruising or striding (40.89 ± 7.04 m·min^{−1}, 95% CI: 37.98–43.79 m·min^{−1} vs. 28.65 ± 5.68 m·min^{−1}, 95% CI: 26.73–30.57 m·min^{−1}, *p* ≤ 0.001, ES = −1.95 [large]), and high-intensity running (19.21 ± 3.08 m·min^{−1}, 95% CI: 17.94–20.48 m·min^{−1} vs. 16.75 ± 4.26 m·min^{−1}, 95% CI: 15.31–18.19 m·min^{−1}, *p* = 0.016, ES = −0.64 [moderate]) compared with the sub-elite midfielder players. Elite midfield players also elicited a significantly higher average speed (8.22 ± 0.64 km·h^{−1}, 95% CI: 7.96–8.49 km·h^{−1} vs. 7.35 ± 0.65 km·h^{−1}, 95% CI: 7.13–7.57 km·h^{−1}, *p* ≤ 0.001, ES = −1.36 [large]) compared with the sub-elite midfielder players. The sub-elite midfield players covered more relative distance walking (42.2 ± 2.63 m·min^{−1}, 95% CI: 41.31–43.09 m·min^{−1} vs. 34.68 ± 3.7 m·min^{−1}, 95% CI: 33.16–36.21 m·min^{−1}, *p* ≤ 0.001, ES = 2.75 [very large]) and sprinting (7.54 ± 2.39 m·min^{−1}, 95% CI: 6.73–8.35 m·min^{−1} vs. 4.23 ± 1.8 m·min^{−1}, 95% CI: 3.48–4.97 m·min^{−1}, *p* ≤ 0.001, ES = 1.52 [large]) than the elite midfield players.

Elite half forwards covered significantly more relative distance jogging (30.91 ± 7.09 m·min^{−1}, 95% CI: 28.67–33.15 m·min^{−1} vs. 26.97 ± 5.12 m·min^{−1}, 95% CI: 25.55–28.4 m·min^{−1}, *p* = 0.004, ES = −0.71 [moderate]) compared with the sub-elite half forwards. However, sub-elite half forwards covered significantly more relative distance walking (40.8 ± 4.01 m·min^{−1}, 95% CI: 39.66–41.94 m·min^{−1} vs. 36.66 ± 4.7 m·min^{−1}, 95% CI: 35.17–38.14 m·min^{−1}, *p* ≤ 0.001, ES = 0.95 [moderate]) than their elite counterparts. All other comparisons between elite and sub-elite half forwards were found to be insignificant.

Elite full forwards covered significantly more relative distance jogging (24.17 ± 5.02 m·min^{−1}, 95% CI: 22.45–25.89 m·min^{−1} vs. 21.47 ± 4.02 m·min^{−1}, 95% CI: 20.67–22.68 m·min^{−1}, *p* = 0.009, ES = −0.6 [moderate]) and elicited significantly longer power events average time (8.15 ± 0.42 seconds, 95% CI: 8–8.29 seconds vs. 7.74 ± 1 seconds, 95% CI: 7.44–8.04 seconds, *p* = 0.017, ES = −0.62 [moderate]) compared with the sub-elite full forwards. Conversely, the sub-elite full forwards covered significantly more relative distance walking (41.44 ± 2.88 m·min^{−1}, 95% CI: 40.58–42.31 m·min^{−1} vs. 37.58 ± 2.12 m·min^{−1}, 95% CI: 36.85–38.31 m·min^{−1}, *p* ≤ 0.001, ES = 1.51 [large]) compared with the elite full forwards.

Playing Level Comparisons

Descriptive statistics for the metrics measured within each team across quarters are displayed in Table 4 as mean ± SD and CV%. The elite team covered significantly greater relative distance (*p* = 0.021, ES = 0.007 [null]) in quarter 2 (115.35 ± 17.7 m·min^{−1}, 95% CI: 112.64–118.05 m·min^{−1} vs. 111.62 ± 19.43 m·min^{−1}, 95% CI: 109.34–113.89 m·min^{−1}, *p* = 0.043, ES = 0.009 [null]), quarter 3 (116.04 ± 22.03 m·min^{−1}, 95% CI: 112.68–119.41 m·min^{−1} vs. 110.79 ± 21.35 m·min^{−1}, 95% CI: 108.29–113.3

Table 3

Acceleration, negative acceleration, speed, velocity, and power event metrics of Gaelic football players in regard to position.*†

Performance variable	Full back (n = 84)				Half back (n = 155)				Midfield (n = 61)				Half forward (n = 93)				Full forward (n = 80)			
	Elite	Sub-elite	p	ES	Elite	Sub-elite	p	ES	Elite	Sub-elite	p	ES	Elite	Sub-elite	p	ES	Elite	Sub-elite	p	ES
Average speed (km·h ⁻¹)	6.32 ± 0.56 (8.86%)	5.72 ± 0.83‡ (14.51%)	≤0.001	−0.01 (trivial)	7.18 ± 0.81 (10.37%)	7.16 ± 0.87 (12.15%)	0.873	−0.03 (trivial)	8.22 ± 0.64 (7.79%)	7.35 ± 0.65‡ (8.84%)	≤0.001	1.36 (large)	7.51 ± 0.83 (11.05%)	7.49 ± 0.88 (11.75%)	0.92	−0.02 (trivial)	5.96 ± 0.67 (11.24%)	6.03 ± 0.63 (10.45%)	0.636	0.11 (trivial)
Maximal velocity (km·h ⁻¹)	29.26 ± 1.42 (4.85%)	29.12 ± 1.52 (1.79%)	0.692	−0.09 (trivial)	29.28 ± 1.94 (6.63%)	28.87 ± 1.57 (5.44%)	0.161	−0.25 (small)	27.45 ± 0.94 (3.42%)	29.3 ± 1.39‡ (4.74%)	≤0.001	1.50 (large)	28.91 ± 1.79 (6.19%)	28.39 ± 1.41 (4.97%)	0.122	−0.33 (small)	29.57 ± 1.48 (5%)	28.88 ± 1.29 (4.47%)	0.03	−0.50 (small)
Maximal acceleration (m·s ⁻²)	4.48 ± 0.40 (8.93%)	4.36 ± 0.34 (7.8%)	0.146	−0.32 (small)	4.30 ± 0.57 (13.26%)	4.1 ± 0.40‡ (9.76%)	0.016	−0.46 (small)	4.11 ± 0.40 (9.73%)	4.16 ± 0.41 (9.86%)	0.627	0.13 (trivial)	4.25 ± 0.56 (13.18%)	3.91 ± 0.36‡ (9.21%)	0.001	−0.84 (moderate)	4.39 ± 0.37 (8.43%)	4.16 ± 0.35‡ (8.41%)	0.006	−0.64 (moderate)
Maximal negative acceleration (m·s ⁻²)	−5.57 ± 0.53 (9.52%)	−5.63 ± 0.79 (14.03%)	0.652	−0.11 (trivial)	−5.93 ± 0.87 (14.67%)	−5.47 ± 0.67‡ (12.25%)	≤0.001	0.66 (moderate)	−5.36 ± 0.73 (13.62%)	−5.65 ± 0.67 (11.86%)	0.113	0.42 (small)	−5.54 ± 0.70 (12.64%)	−5.48 ± 0.55 (10.04%)	0.645	0.10 (trivial)	−5.82 ± 0.68 (11.68%)	−5.65 ± 0.88 (15.58%)	0.341	0.22 (small)
Power events per minute (n)	1.40 ± 0.22 (15.71%)	1.19 ± 0.30‡ (25.21%)	0.001	−0.88 (moderate)	1.59 ± 0.27 (16.98%)	1.65 ± 0.26 (15.75%)	0.146	0.24 (small)	1.81 ± 0.12 (6.63%)	1.74 ± 0.27 (15.52%)	0.167	0.39 (small)	1.61 ± 0.21 (13.04%)	1.63 ± 0.29 (17.79%)	0.731	0.07 (trivial)	1.21 ± 0.21 (17.35%)	1.18 ± 0.16 (13.56%)	0.589	−0.12 (trivial)
Power events average time (s)	7.49 ± 0.38 (5.07%)	7.57 ± 0.63 (8.32%)	0.469	0.18 (trivial)	8.05 ± 0.56 (6.96%)	8.29 ± 0.82‡ (9.89%)	0.038	0.34 (small)	8.31 ± 0.52 (6.26%)	8.43 ± 1.12 (13.29%)	0.599	0.15 (trivial)	8.44 ± 0.60 (7.11%)	8.19 ± 0.86 (10.5%)	0.099	−0.35 (small)	8.15 ± 0.42 (5.15%)	7.74 ± 1.00‡ (12.92%)	0.017	−0.62 (moderate)

*CV% = coefficient of variation percentage.

†Data are presented as mean ± SD (CV%).

‡Significantly different (p < 0.05) from elite level; power events are defined as the estimation of the number of events in a session, which depend on anaerobic processes, including sprints, accelerations, and changes of direction.

m·min⁻¹, p = 0.013, ES = 0.014 [small]), and quarter 4 (118.05 ± 20.29 m·min⁻¹, 95% CI: 114.95–121.15 m·min⁻¹ vs. 110.97 ± 20.29 m·min⁻¹, 95% CI: 108.63–113.3 m·min⁻¹, p ≤ 0.001, ES = 0.028 [small]; Figure 1). The elite team also performed significantly more relative high-intensity running (p = 0.002, ES = 0.012 [small]) in quarter 3 (15.28 ± 5.05 m·min⁻¹, 95% CI: 14.49–16.08 m·min⁻¹ vs. 13.96 ± 6.12 m·min⁻¹, 95% CI: 13.23–14.7 m·min⁻¹, p = 0.023, ES = 0.012 [small]) compared with the sub-elite team (Figure 2). The elite team also had a significantly higher average speed (p = 0.023, ES = 0.007 [null]) in quarter 2 (6.92 ± 1.06 km·h⁻¹, 95% CI: 6.56–6.84 km·h⁻¹ vs. 6.7 ± 1.12 km·h⁻¹, 95% CI: 6.52–6.83 km·h⁻¹, p = 0.048, ES = 0.009 [null]), quarter 3 (6.97 ± 1.14, 95% CI: 6.79–7.14 km·h⁻¹ vs. 6.67 ± 1.32 km·h⁻¹, 95% CI: 6.52–6.83 km·h⁻¹, p = 0.017, ES = 0.013 [small]), and quarter 4 (7.09 ± 1.14 km·h⁻¹, 95% CI: 6.92–7.27 km·h⁻¹ vs. 6.65 ± 1.19 km·h⁻¹, 95% CI: 6.51–6.79 km·h⁻¹, p ≤ 0.001, ES = 0.032 [small]) compared with the sub-elite team (Figure 3). Significantly more power events per minute (p = 0.006, ES = 0.009 [null]) were also seen in the elite cohort in quarter 4 (1.53 ± 0.39 power events, 95% CI: 1.47–1.59 power events vs. 1.44 ± 0.41 power events, 95% CI: 1.39–1.48 power events, p = 0.015, ES = 0.013 [small]) than in the sub-elite cohort; however, the sub-elite team performed significantly more power events per minute (p = 0.006, ES = 0.009 [null]) in quarter 1 (1.62 ± 0.45 power events, 95% CI: 1.57–1.67 vs. 1.52 ± 0.38 power events, 95% CI: 1.46–1.58 power events, p = 0.020, ES = 0.012 [small]) than the elite team (Figure 4). The sub-elite team also performed significantly more relative high-speed running (p = 0.011, ES = 0.009 [null]) in quarter 1 (23.89 ± 8.49 m·min⁻¹, 95% CI: 22.88–24.9 m·min⁻¹ vs. 20.47 ± 7.2 m·min⁻¹, 95% CI: 19.29–21.66 m·min⁻¹, p ≤ 0.001, ES = 0.039 [small]; Figure 5) and relative high-intensity running (p = 0.002, ES = 0.012 [small]) in quarter 1 (16.37 ± 6.36 m·min⁻¹, 95% CI: 15.6–17.13 m·min⁻¹ vs. 14.54 ± 5.43 m·min⁻¹, 95% CI: 13.69–15.4 m·min⁻¹, p = 0.003, ES = 0.021 [small]) compared with the elite team (Figures 2 and 6).

Discussion

The aim of the current study was to analyze the differences in match-play running performance between elite and sub-elite Gaelic football players. To account for the discrepancy in match duration between elite and sub-elite match-play, total running distance, high-speed running distance, distance within each speed zone, and power events were analyzed and reported in relative terms (per minute). The running performance variables analyzed were also analyzed by quarter of match-play to allow for analysis of trends between each team's performance. Differences were expected as previous research in Rugby League (n = 100; 36 elite and 64 sub-elite) and Australian football (n = 41; 15 elite, 8 sub-elite, and 18 who participated at both levels) consistently reported differences between elite and sub-elite levels (2,3).

Significant differences were present on a team and positional level. Several of the differences presented within this study seemed to be because of the superior conditioning of the elite-level players as a result of increased preparation, which may result in the elite players experiencing less fatigue in comparison with their sub-elite counterparts (37). This is evident in the results in relation to average speed, relative total distance, high-speed running, high-intensity running, and sprinting. Also, relative total distance, high-intensity running, high-speed running, and average speed

scores decreased within the sub-elite players while the elite scores remain relatively consistent as the game progressed, which may be indicative of the sub-elite players beginning the match at a pace that they are unable to maintain (Table 4). This is also evident in the results regarding power events per minute.

Elite full backs and midfielders ran significantly more relative total distance compared with their sub-elite counterparts. These results are in agreement with the results seen in hurlers by Young et al. (38). Relative total distance among elite full backs was similar in the present study ($105.16 \pm 9.23 \text{ m} \cdot \text{min}^{-1}$) as has been reported in elite Gaelic football players previously ($116 \pm 21 \text{ m} \cdot \text{min}^{-1}$ to $118 \pm 9 \text{ m} \cdot \text{min}^{-1}$) (15,16). Relative total distance within sub-elite full backs was also similar in the present study ($95.3 \pm 13.81 \text{ m} \cdot \text{min}^{-1}$) compared with previous research in sub-elite Gaelic footballers (95.1 ± 14.5) (19). When analyzing on a team level, the elite footballers covered more relative distance than the sub-elite footballers in quarter 2, 3, and 4. This finding has a level of agreement with those of Young et al. (38), who found that elite hurlers to cover more relative total distance in the first ($108 \pm 9 \text{ m} \cdot \text{min}^{-1}$ vs. $96 \pm 17 \text{ m} \cdot \text{min}^{-1}$) and second half ($115 \pm 10 \text{ m} \cdot \text{min}^{-1}$ vs. $90 \pm 16 \text{ m} \cdot \text{min}^{-1}$) of match-play compared with the sub-elite-level players. When analyzing full backs and midfielders, the elite players also elicited a significantly higher average speed than their sub-elite counterparts. On a team level, the elite team also elicited a higher average speed than the sub-elite team in the second, third and fourth quarters. Although, the effect size for each of these quarters was found to be small.

The sub-elite players performed significantly more relative high-speed running, high-intensity running, and power events per minute in quarter 1. This only occurs in the first quarter and no others, and the sub-elite team experience a decrease in relative high-speed and high-intensity running and power events per minute after quarter 1 (13.23, 13.44, and 11.11% decrement, respectively, after quarter 1), adding to the theory that the sub-elite players begin at a pace that they can not maintain. However, the effect sizes for these results regarding relative high-speed running, relative high-intensity running, and power events per minute were null, small, and small, respectively. Mohr et al. (24) reported entries into the high-speed running zone ($18 \text{ km} \cdot \text{h}^{-1}$) in professional soccer players who were considered either top-class (who played for an elite European team who partook in the Italian and European Champions League) or moderate ability (who played for a top Danish league team). This study found the top-class players to have more entries than the moderate ability players (69 ± 5 vs. 49 ± 3) (24). Conversely, within the study of Young et al. (38) in hurlers, there was no significant difference between the elite and sub-elite players in high-speed running within the first half. Potentially, these high-speed running scores presented in hurling are significantly different to those presented in the current study because, although hurling and Gaelic Football are similar in pitch size, goal size, and number of players, within the game of hurling, there is potential for playing the ball (sliotar) longer distances down the pitch, lessening the running demands and the need for high-speed running. In the current study, the high-speed running difference was significant in quarter 1 ($p < 0.001$) between levels but insignificant in quarter 2 ($p = 0.337$). The elite team performed significantly more high-intensity running in quarter 3 and power events per minute in quarter 4. This serves to further the argument that the sub-elite players begin the match at a pace that they can not maintain as the elite team presented relatively consistent high-intensity running and power events per minute values within each quarter of match play. Because of the conflicting research across sports, more

research analyzing match-quarters and a larger cohort, consisting of more than 2 teams would be needed to draw a more concrete conclusion from these findings. The results of this study and those of other studies may also differ because of playing standards of the teams being analyzed, tactical approaches to the game, the demands of each sport, league position, and performance of each team throughout the season, which may affect running demands and fitness levels.

Elite midfielders performed significantly more high-intensity running than the sub-elite midfielders. The sub-elite midfielders' high-intensity values in the present study ($16.75 \pm 4.26 \text{ m} \cdot \text{min}^{-1}$) are similar to those previously recorded ($15.2 \pm 4.1 \text{ m} \cdot \text{min}^{-1}$) (19). This point may be more evident when comparing elite and sub-elite midfielders than in other positions because elite midfielders have been shown to perform the most total and high-speed running compared with all other positions, showing that the running demands within this position are among the highest within elite Gaelic football and hurling (16,23,38). The elite midfield players within the present study performed significantly more high-intensity running ($19.21 \pm 3.08 \text{ m} \cdot \text{min}^{-1}$) than has been previously presented in elite hurlers ($2.9 \pm 1.1 \text{ m} \cdot \text{min}^{-1}$) (38). Hurling and Gaelic football have similar pitch sizes and number of players on the field; however, some running demands may be different because of the nature of the game and the ability of the hurlers to play the ball longer distances downfield, reducing their need for high-intensity running. This trend of elite players performing more high-speed running than lower-level players has also been seen within professional soccer players (24).

Sub-elite full backs, half backs, midfielders, half forwards, and full forwards covered more distance walking and sub-elite full backs and midfielders covered more distance sprinting compared with the elite counterparts. These results mirror those of Young et al. (38). It may be speculated that the sub-elite players' inability to read the game resulted in them performing more sprinting to make up ground and more walking after these sprints because of their lower fitness levels in comparison with the elite-level players. Young et al. (38) suggested that the sub-elite players may be unable to withstand the demands of elite hurling because of a lack of conditioning, resulting in them performing more walking and more sprinting to make up ground. This is further echoed in the current study because the sub-elite half backs performed significantly more relative high-speed running and the sub-elite full forwards performed significantly more relative high-speed running within quarter 1 compared with the elite counterparts, compounded by the sub-elite players experiencing a reduction in relative high-speed running ($\geq 17 \text{ km} \cdot \text{h}^{-1}$) distance and relative high-intensity running ($17\text{--}21.9 \text{ km} \cdot \text{h}^{-1}$) after quarter 1 (13.23 and 13.44% decrement, respectively, after quarter 1). This can be seen in the data in this study because there is a drop in relative high-speed running distance in the sub-elite players after quarter 1 ($23.89 \pm 8.49 \text{ m} \cdot \text{min}^{-1}$ in quarter 1 and $20.73 \pm 7.89 \text{ m} \cdot \text{min}^{-1}$ in quarter 2), whereas the elite players remained relatively consistent ($20.47 \pm 7.2 \text{ m} \cdot \text{min}^{-1}$ in quarter 1 and $20.00 \pm 6.33 \text{ m} \cdot \text{min}^{-1}$ in quarter 2). These trends regarding high-speed running could point toward the sub-elite players' inability to pace themselves compared with the elite players. This can also be seen as the sub-elite players performed significantly higher maximum negative acceleration in quarters 1 and 3 (Figure 6). It could be speculated that the sub-elite players elicit higher maximum negative acceleration, in order for them to change directions after misreading a play. However, the effect size for these metrics were also null and small, respectively. Also, GPS has been shown to provide inconsistent findings for measuring velocity and maximal

Table 4

Running performance variables of Gaelic football players in regard to position.*†

Performance variable	Standing and walking (0–6.9 km·h ⁻¹) (m·min ⁻¹)		Jogging (7–11.9 km·h ⁻¹) (m·min ⁻¹)		Cruising or striding (12–16.9 km·h ⁻¹) (m·min ⁻¹)		High-intensity running (17–21.9 km·h ⁻¹) (m·min ⁻¹)		Sprinting (>22 km·h ⁻¹) (m·min ⁻¹)	
	Elite	Sub-elite	Elite	Sub-elite	Elite	Sub-elite	Elite	Sub-elite	Elite	Sub-elite
Quarter 1	37.16 ± 3.98 (10.71%)	41.30 ± 4.05 (9.81%)	30.93 ± 8.34 (26.94%)	26.69 ± 7.31 (27.39%)	29.42 ± 11.49 (39.06%)	27.17 ± 10.43 (38.39%)	14.54 ± 5.43 (37.35%)	16.37 ± 6.36‡ (38.85%)	5.92 ± 3.11 (52.53%)	7.04 ± 3.62 (51.42%)
Quarter 2	38.47 ± 3.76 (9.77%)	42.73 ± 4.37 (10.23%)	29.39 ± 7.56 (25.72%)	23.6 ± 6.46 (27.37%)	27.07 ± 8.47 (31.29%)	23.05 ± 8.32 (36.1%)	13.95 ± 4.88 (34.98%)	14.17 ± 5.84 (41.21%)	5.60 ± 2.99 (53.39%)	5.74 ± 2.95 (51.39%)
Quarter 3	37.59 ± 4.61, (12.26%)	41.29 ± 4.12 (9.98%)	27.88 ± 7.56 (27.12%)	23.45 ± 6.50 (27.72%)	27.49 ± 10.12 (36.81%)	24.51 ± 9.48 (38.68%)	15.28 ± 5.05 (33.05%)	13.96 ± 6.12‡ (43.84%)	6.21 ± 3.19 (51.37%)	6.37 ± 3.47 (53.53%)
Quarter 4	37.7 ± 4.33 (11.49%)	41.85 ± 5.08 (12.14%)	28.36 ± 6.66 (23.48%)	22.75 ± 5.94 (26.11%)	29.40 ± 10.03 (34.12%)	23.93 ± 8.64 (36.11%)	14.84 ± 5.98 (40.3%)	14.65 ± 6.83 (46.62%)	5.23 ± 3.20 (61.19%)	5.85 ± 3.56 (60.85%)
<i>p</i>		0.799		0.257		0.083		0.002		0.106
ES		0.001		0.003		0.005		0.012		0.005

Performance variable	Relative distance (m·min ⁻¹)		Relative high-speed running (m·min ⁻¹)		Average speed (km·h ⁻¹)		Maximal velocity (km·h ⁻¹)		Maximal acceleration (m·s ⁻²)	
	Elite	Sub-elite	Elite	Sub-elite	Elite	Sub-elite	Elite	Sub-elite	Elite	Sub-elite
Elite	Sub-elite	Elite	Sub-elite	Elite	Sub-elite	Elite	Sub-elite	Sub-elite	Elite	Sub-elite
Quarter 1	118.52 ± 21.51 (18.15%)	119.69 ± 21.06 (17.6%)	20.47 ± 7.2 (35.17%)	23.89 ± 8.49‡ (35.54%)	7.13 ± 1.29 (18.09%)	7.18 ± 1.26 (17.55%)	27.9 ± 2.05 (7.35%)	27.7 ± 1.87 (6.75%)	3.96 ± 0.53 (13.38%)	3.81 ± 0.45 (11.81%)
Quarter 2	115.35 ± 17.7 (15.34%)	111.62 ± 19.43‡ (17.41%)	20.00 ± 6.33 (31.65%)	20.73 ± 7.89 (38.06%)	6.92 ± 1.06 (15.32%)	6.7 ± 1.12‡ (16.72%)	27.51 ± 2.06 (7.49%)	27.39 ± 1.89 (6.9%)	3.78 ± 0.45 (11.9%)	3.77 ± 0.48 (12.73%)
Quarter 3	116.04 ± 22.03 (18.98%)	110.79 ± 21.35‡ (19.27%)	21.08 ± 5.05 (23.96%)	21.15 ± 8.68 (41.04%)	6.97 ± 1.14 (16.36%)	6.67 ± 1.32‡ (19.79%)	27.53 ± 1.92 (6.97%)	27.58 ± 1.84 (6.67%)	3.94 ± 0.49 (12.44%)	3.76 ± 0.48 (12.77%)
Quarter 4	118.05 ± 20.29 (17.19%)	110.97 ± 20.29‡ (18.28%)	19.77 ± 7.16 (35.22%)	20.51 ± 8.75 (42.66%)	7.09 ± 1.14 (16.08%)	6.65 ± 1.19‡ (17.89%)	26.92 ± 2.28 (8.47%)	27.1 ± 1.98 (7.31%)	3.82 ± 0.60 (15.71%)	3.70 ± 0.47 (12.7%)
<i>p</i>		0.021		0.011		0.023		0.512		0.065
ES		0.007		0.009		0.007		0.002		0.006

Performance variable	Maximal negative acceleration (m·s ⁻²)		Power events per minute (n)		Power events average time (s)	
	Elite	Sub-elite	Elite	Sub-elite	Elite	Sub-elite
Quarter 1	–5.23 ± 0.91 (17.4%)	–4.91 ± 0.79‡ (16.09%)	1.52 ± 0.38 (25%)	1.62 ± 0.45‡ (27.7%)	8.10 ± 0.96 (11.85%)	8.08 ± 1.12 (13.86%)
Quarter 2	–4.85 ± 0.68 (14.02%)	–4.86 ± 0.85 (17.49%)	1.45 ± 0.32 (22.07%)	1.44 ± 0.41 (28.47%)	8.06 ± 1.01 (12.53%)	8.07 ± 1.28 (15.86%)
Quarter 3	–5.04 ± 0.89 (17.66%)	–4.7 ± 0.86‡ (18.3%)	1.49 ± 0.38 (25.5%)	1.48 ± 0.44 (29.73%)	8.25 ± 0.85 (10.3%)	8.15 ± 1.41 (17.3%)
Quarter 4	–4.72 ± 0.84 (17.8%)	–4.66 ± 0.81 (17.38%)	1.53 ± 0.39 (25.49%)	1.44 ± 0.41‡ (28.47%)	8.14 ± 1.16 (14.25%)	8.09 ± 1.22 (15.08%)
<i>p</i>		0.003		0.006		0.912
ES		0.011		0.009		≤0.001

*CV% = coefficient of variation percentage.

†Data are presented as mean ± SD (CV%).

‡Significantly different (*p* < 0.05) from elite level; power events are defined as the estimation of the number of events in a session, which depend on anaerobic processes, including sprints, accelerations, and changes of direction.

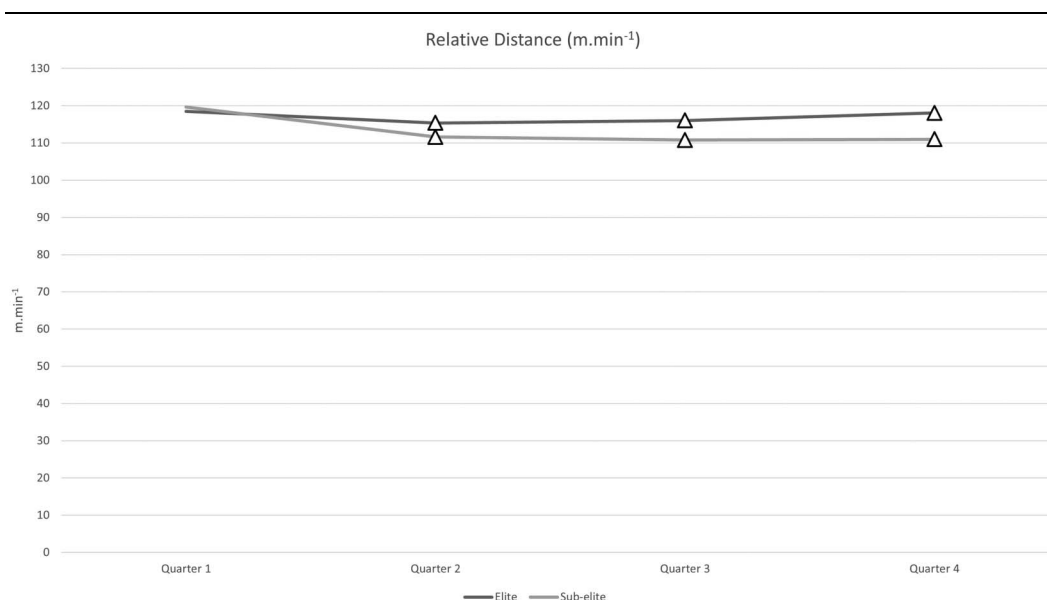


Figure 1. Relative distance ($\text{m}\cdot\text{min}^{-1}$) within elite and sub-elite players across 4 quarters of match-play. $\triangle$ Significantly different ($p < 0.05$).

acceleration data (12). Elite full backs, half backs, midfielders, half forwards, and full forwards performed significantly more relative jogging than their sub-elite counterparts, which may indicate the elite players' ability to predict plays better than the sub-elite players, lessening the need for high-speed and high-intensity running and sprinting.

The elite full backs and midfielders also covered significantly more distance cruising or striding than the sub-elite counterparts. McGahan et al. (22) speculated that division 3 elite Gaelic football teams may perform more hand passes when maneuvering the ball down the field and division 1 elite teams perform more kick passes. As a result, the division 1 players had decreased running demands compared with division 3 players, which may be the case between the elite and sub-elite teams in the current study. As such,

it may be a combination of the elite players' use of kick passing and ability to read the game and anticipate plays that resulted in more time spent jogging, and cruising or striding for the elite players and the sub-elites performing significantly more high-intensity running, sprinting, and subsequently, walking, possibly because of fatigue (43). This may also explain the sub-elite half backs having significantly longer power events average time and the sub-elite midfielders eliciting a higher maximal velocity compared with the elite players. The sub-elite players may have to sprint for longer periods, at higher velocities, to make up ground because of their inability to read the game and ineffective use of kick passing. The maximal velocity elicited by the sub-elite midfielders within the current study ($29.3 \pm 1.39 \text{ km}\cdot\text{h}^{-1}$) is also similar to that reported in previous research in sub-elite midfield

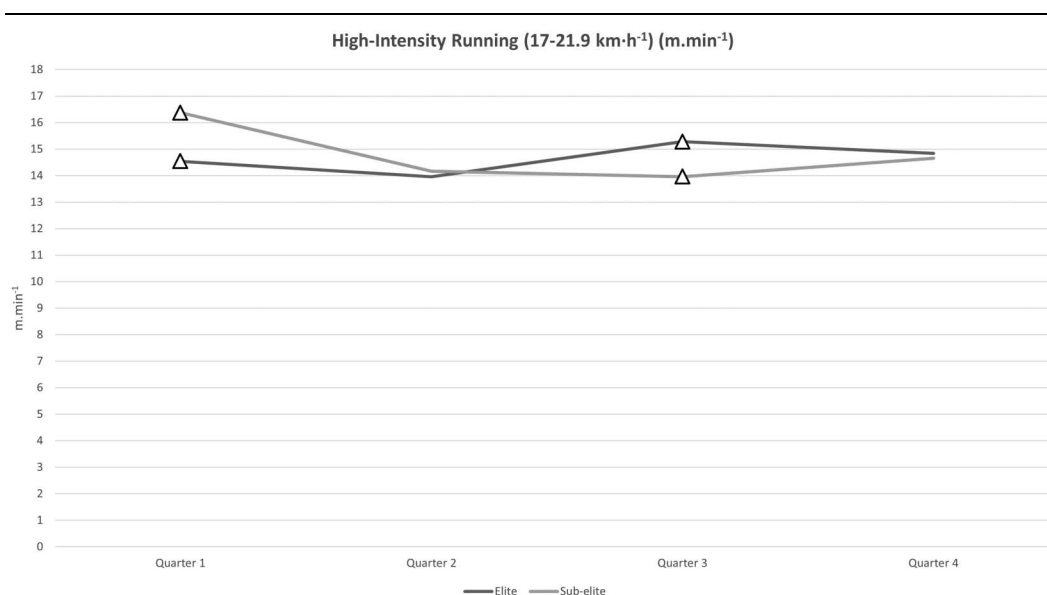


Figure 2. High-intensity running ($17\text{--}21.9 \text{ km}\cdot\text{h}^{-1}$) ($\text{m}\cdot\text{min}^{-1}$) within elite and sub-elite players across 4 quarters of match-play. $\triangle$ Significantly different ($p < 0.05$).

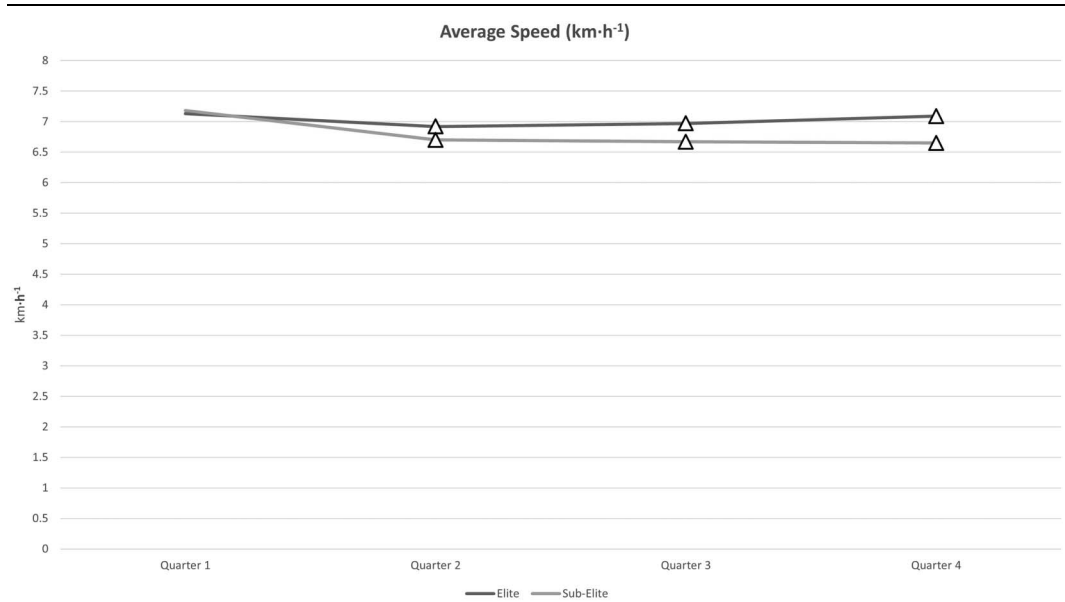


Figure 3. Average speed ($\text{km}\cdot\text{h}^{-1}$) within elite and sub-elite players across 4 quarters of match-play. Δ Significantly different ($p < 0.05$).

Gaelic football players ($28.8 \pm 1.7 \text{ km}\cdot\text{h}^{-1}$) (19). This trend of sub-elite eliciting a larger maximal velocity than elite has also been seen in hurlers (38). However, the maximal velocity of the elite midfielders within the present study ($27.45 \pm 0.94 \text{ km}\cdot\text{h}^{-1}$) is below that found in elite hurlers ($29.8 \pm 2.3 \text{ km}\cdot\text{h}^{-1}$), which may exacerbate the effect size of this difference within the present study ($\text{ES} = 1.19$ [moderate]) (45).

The elite full backs performed more power events per minute and the elite full forwards elicited a higher maximum velocity and averaged a longer power event duration than the sub-elite players, which may be indicative of the increased demand on anaerobic processes within these positions in elite Gaelic football in comparison with sub-elite players, which has been previously seen in

soccer (33). Maximum velocity values within the present study for full forwards ($29.57 \pm 1.48 \text{ km}\cdot\text{h}^{-1}$) are similar to those previously sighted in elite hurling ($29.6 \pm 2.2 \text{ km}\cdot\text{h}^{-1}$) (5). Further adding to this, elite half backs, half forwards, and full forwards had a larger maximum acceleration, and the elite half backs elicited a larger maximal negative acceleration than the sub-elite players. Additionally, similar maximum acceleration ($\text{m}\cdot\text{s}^{-2}$) values were seen in the current study in the sub-elite half backs ($4.1 \pm 0.4 \text{ m}\cdot\text{s}^{-2}$), half forwards ($3.91 \pm 0.36 \text{ m}\cdot\text{s}^{-2}$), and full forwards ($4.16 \pm 0.35 \text{ m}\cdot\text{s}^{-2}$) as have been seen in sub-elite players in previous research (4.3 ± 0.3 , 4.1 ± 0.4 , and $4.2 \pm 0.4 \text{ m}\cdot\text{s}^{-2}$, respectively) (19). Maximal negative acceleration values within the present study for sub-elite half backs (-5.47 ± 0.67

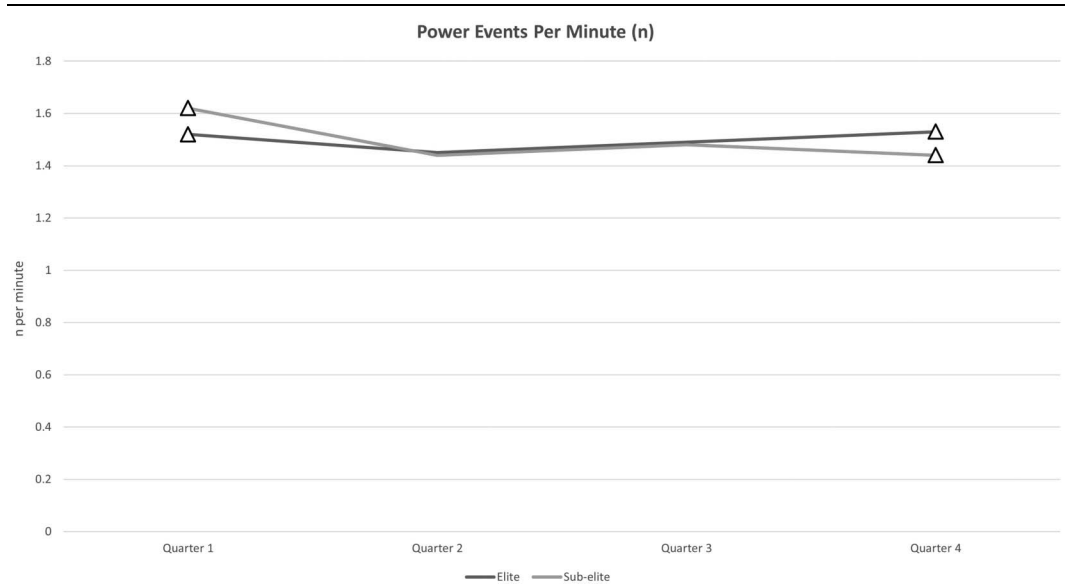


Figure 4. Power events per minute (n) within elite and sub-elite players across 4 quarters of match-play. Δ Significantly different ($p < 0.05$); power events are defined as the estimation of the number of events in a session, which depend on anaerobic processes, including sprints, accelerations, and changes of direction.

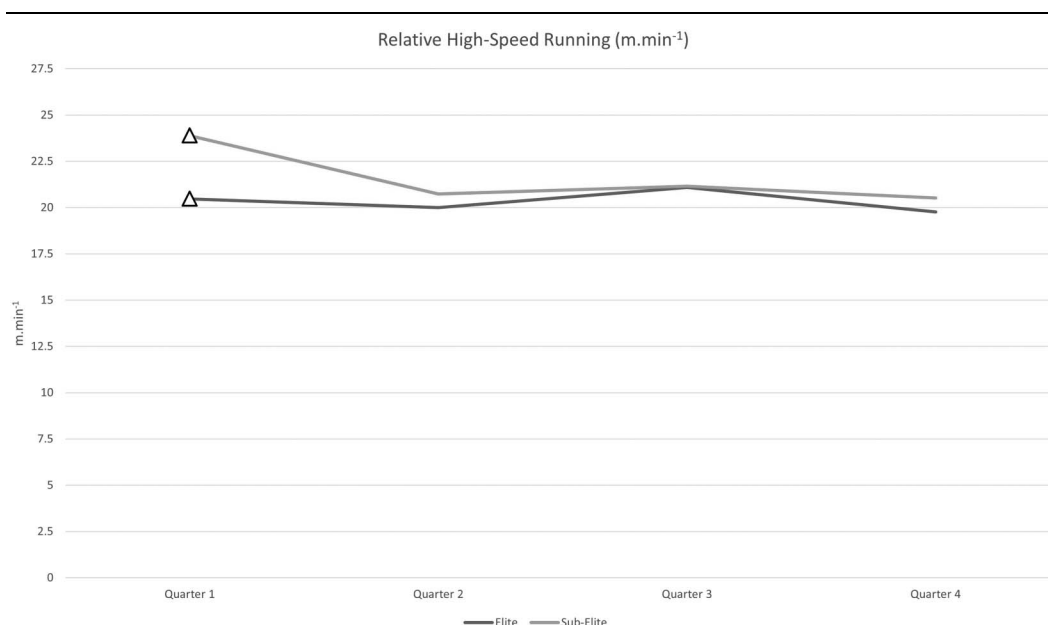


Figure 5. Relative high-speed running ($\text{m}\cdot\text{min}^{-1}$) within elite and sub-elite players across 4 quarters of match-play. $\triangle$ Significantly different ($p < 0.05$).

$\text{m}\cdot\text{s}^{-2}$) and full forwards ($-5.65 \pm 0.88 \text{ m}\cdot\text{s}^{-2}$) are also similar to those previously found in sub-elite half backs ($-5.7 \pm 0.7 \text{ m}\cdot\text{s}^{-2}$) and full forwards ($-6 \pm 0.7 \text{ m}\cdot\text{s}^{-2}$) (19). Additionally, in the current study, no statistical differences were found between teams for walking ($p = 0.799$), jogging ($p = 0.257$), cruising or striding ($p = 0.083$), sprinting ($p = 0.106$), maximal velocity ($p = 0.512$), maximal accelerations ($p = 0.065$), and power events average time ($p = 0.912$).

The current study contains several limitations. First, 27 sub-elite matches were analyzed, and only 16 elite matches were analyzed (59% of the sub-elite matches). This may lead to the study

having a less accurate representation of the elite cohort compared with the sub-elite cohort. This occurs as sub-elite teams have more games per season, particularly if they are successful. Also, only one team from each level of play was examined; therefore, the results of this study are specific to these teams and may not be fully representative of all elite and sub-elite teams. Additionally, no data concerning heart rate, oxygen uptake ($\dot{V}\text{O}_2$), or any tactical variables were assessed. Heart rate has previously been shown to vary in elite and sub-elite hurlers (38), rugby union players (9), and Australian football players (3). $\dot{V}\text{O}_2$ has also been found to vary between elite and sub-elite soccer players (36). As

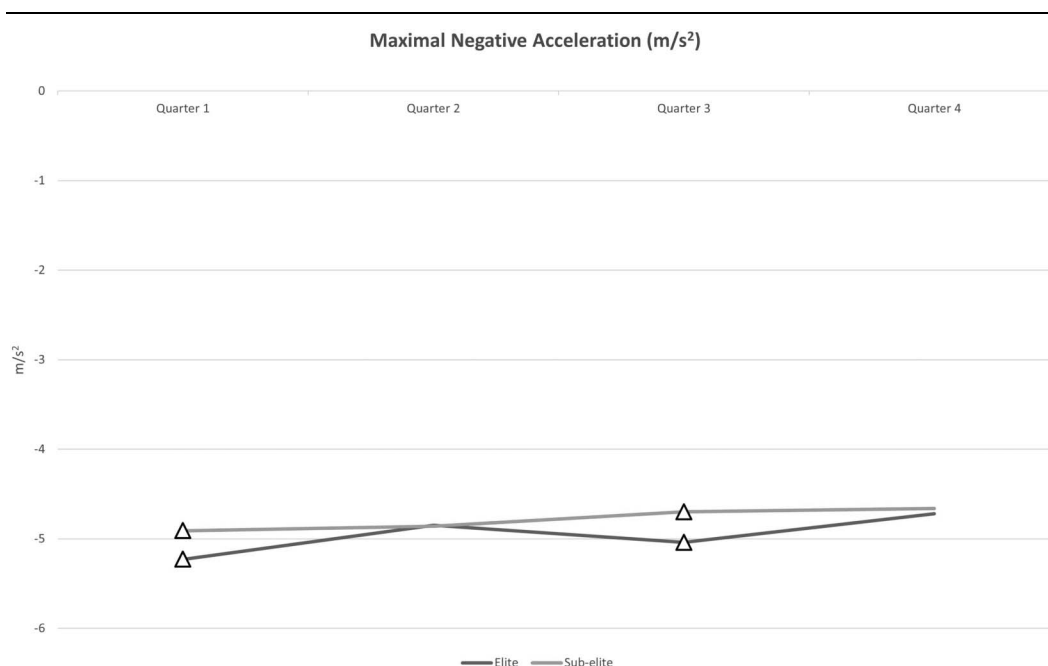


Figure 6. Maximal negative acceleration ($\text{m}\cdot\text{s}^{-2}$) within elite and sub-elite players across 4 quarters of match-play. $\triangle$ Significantly different ($p < 0.05$); negative numbers representing meters per second decrease in speed.

such, future research should examine heart rate, $\dot{V}O_2$, and tactical variables to assess potential differences within these metrics between elite and sub-elite players. Additionally, goalkeepers were not analyzed as part of the research. As such, the results of this study are only applicable to outfield players, and further research is needed analyzing match-play performance within goalkeepers. The discrepancy in match duration between the elite and sub-elite cohorts is also a limitation when comparing the 2 levels, despite the running variables being converted into relative terms. It could be speculated that if the elite players were to compete in a 60-minute match, they would have higher relative scores because they would not have the pace themselves as much compared with a 70-minute match. Additionally, many Gaelic football players compete at both sub-elite level and elite level simultaneously, which may influence results when attempting to compare elite and sub-elite Gaelic football players because some players who are labelled as sub-elite within the data may also be elite players. Also, players were described using their positional line at the start of each match, which did not account for players changing position during the match. Coefficient of variation (%) is very high in some variables (Table 4), varying between 1.79 and 61.19% across the data, indicating a high variance in values for running performance metrics within positions and teams. However, this is only present within comparisons between quarters where no positional breakdown is present. As such, depending on the style of play of a team or player, or the conditioning of a player, running demands may differ. GPS has previously provided inconsistent validity for measuring power and force data, meaning findings within the current study relating to sprint components may not be applicable (12,25). Future studies may assess GPS data to analyze spacing of the teams during match-play and infer strategies from this information.

Practical Applications

The aim of this study was to examine the differences in match-play running performance between elite and sub-elite Gaelic football players. When drawing comparisons at the team level, no result elicited larger than a small effect size. However, larger differences were seen at the positional level. The results of this study support the initial hypothesis that the conditioning may be superior among elite players in comparison with sub-elite players. Additionally, the elite players may possess a greater ability to read match-play and a larger demand on anaerobic processes may be present within elite Gaelic football. Some of the data regarding total relative distance, high-speed running, high-intensity running, average speed, and power events per minute within quarter 1 also highlight the sub-elite players' inability to pace themselves at the outset of a match. The sub-elite players experience a significant decrease in power events per minute after quarter 1. Given that power events are events that depend on anaerobic processes, the sub-elite players may also consider introducing more anaerobic work, such as sprints, accelerations, and changes of direction, into their training and working on pacing themselves to be able to perform with consistency throughout match-play, which could possibly lead to superior performance in the latter half of match-play. The sub-elite players also perform more walking and more sprinting to make up ground because of lack of conditioning and inefficient use of kick passing, which is further echoed in the sub-elite players' reduction in relative high-speed running (≥ 17

$\text{km}\cdot\text{h}^{-1}$) distance and relative high-intensity running ($17\text{--}21.9 \text{ km}\cdot\text{h}^{-1}$) after quarter 1. The differences presented in this study and potential rationale for these differences may highlight to the strength and conditioning professional, who is working alongside a sub-elite Gaelic football team, where training may be adapted to meet the standard of the elite level. An example of this may include the implementation of small-sided games within training to improve conditioning, which may also improve the ability of the sub-elite players to pace themselves during match-play (20). Using a $60 \times 20\text{-m}$ pitch increases the intensity of the small-sided game above match demands, which could benefit conditioning, and using a $40 \times 20\text{-m}$ pitch has been shown to elicit a greater number of power events, which may increase the anaerobic demands of the game (20). Therefore, using these 2 pitch sizes could benefit the playing standard and conditioning of sub-elite teams. The Yo-Yo intermittent recovery (IR) level 1 and the Yo-Yo IR level 2 may be implemented to test the aerobic and anaerobic systems, respectively, to examine the efficacy of an intervention including either of the above pitch sizes (1).

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